FRESHROUTE - SMART FRUIT SUPPLY CHAIN SYSTEM TO REDUCE WASTE AND ELIMINATE MIDDLEMEN IN SRI LANKA

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Project Proposal Report

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DECLARATION

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by any other person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic, or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

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ABSTRACT

Sri Lanka's fruit supply chain suffers from systemic inefficiencies that reduce farmer incomes and increase post-harvest losses. Studies estimate that 20 to 30 percent of harvested fruits are wasted due to poor harvest planning, reliance on intermediaries, inadequate logistics coordination, and lack of quality assurance mechanisms. The overarching purpose of this study is to design and evaluate a Smart Fruit Supply Chain System that integrates artificial intelligence (AI), blockchain, logistics optimization, and quality verification to address these challenges. Specifically, the research investigates four interlinked problems: inaccurate demand forecasting, middleman dominance, spoilage during transport, and disputes over fruit quality. The proposed system consists of four modules: an AI-based demand forecasting tool and Prophet to guide optimal harvesting schedules, a blockchain marketplace enabling transparent farmer-to-buyer transactions via smart contracts, a spoilage-reduction logistics module that employs dynamic route optimization algorithms with real-time traffic and perishability costs to minimize transit delays and fruit deterioration. The study adopts a mixed-method design, combining field data collection such as fruit prices, yields, transport times, and quality assessments with simulation experiments of the proposed models. By integrating AI forecasting, blockchain-based trust, optimized logistics, and automated quality verification into one platform, the system is expected to reduce post-harvest fruit waste by 20 to 30 percent, enhance farmer revenues by 40 to 50 percent through direct trading, and improve buyer confidence through transparent traceability. This research contributes a novel, data-driven solution to modernize Sri Lanka's fragmented fruit supply chain and provides a scalable model for improving agricultural value chains in similar developing economies.

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
CNN	Convolutional Neural Networks
AI	Artificial Intelligence
IoT	Internet of Things
GPS	Global Positioning System
LSTM	Long Short-Term Memory (neural network type)
VRP	Vehicle Routing Problem
PIRP	Perishable Inventory Routing Problem
NIPHM	National Institute of Post-Harvest Management
API	Application Programming Interface
IDE	Integrated Development Environment

1 INTRODUCTION

Post-harvest fruit losses remain a critical issue in Sri Lanka, with estimates ranging from 20% to 27% of total production being lost before reaching consumers [1], [2]. Transportation stages alone contribute significantly to these inefficiencies. For instance, studies on the banana value chain indicate that 7.6–8% of losses occur specifically during the movement of bananas between wholesale and retail stages, primarily due to poor handling and inappropriate transport practices [2].The drivers of these losses include insufficient cold-chain facilities, inadequate packaging practices, and a fragmented logistics infrastructure that lacks transparency and coordination [3], [4]. Farmers frequently transport fruits in bulk using open trucks or tractors, leading to compression damage, vibration stress, and sun exposure that accelerate spoilage [3]. Smallholder farmers, who dominate Sri Lanka's agricultural landscape, are particularly vulnerable because they often rely on informal transport networks and lack access to reliable cold-chain solutions [6].

Efforts have been made to mitigate these challenges. The National Institute of Post-Harvest Management (NIPHM) has introduced crate distribution programs and farmer training initiatives, but adoption has been limited and not scaled nationwide [5]. Consequently, systemic inefficiencies such as mismatched vehicle allocation, poor packaging, and inadequate cold-chain access continue to drive spoilage. Given these challenges, there is a pressing need for logistics solutions that prioritize freshness preservation rather than focusing solely on travel distance or time.

Figure 1: Spoiled fruits during transportation

While global research has developed optimization models such as the Vehicle Routing Problem (VRP) and the Perishable Inventory Routing Problem (PIRP) to balance cost, routing, and perishability [7], [8], these approaches are largely designed for export-oriented systems in developed economies where cold-chain infrastructure is well established. In contrast, Sri Lanka requires a context-sensitive solution that dynamically assigns vehicles (refrigerated and non-refrigerated) to harvests based on perishability, while also incorporating optimized routing and transport monitoring. This project proposes such an intelligent logistics framework, specifically tailored to Sri Lanka's fruit supply chain, to reduce spoilage and improve efficiency from farm to market.

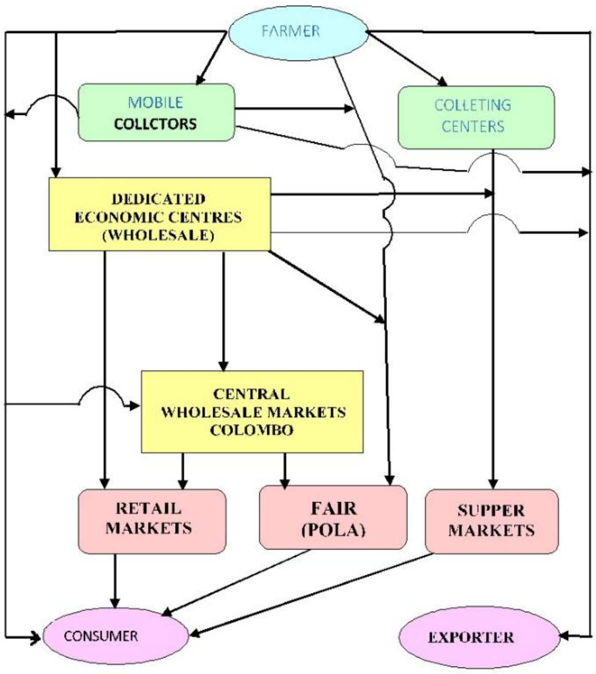

Figure 2: Vegetable and fruit fresh (marketing) channels.

1.1 Background Literature

Sri Lanka's agricultural economy relies heavily on fruit production, which plays a crucial role in farmers' livelihoods and the national food supply. However, post-harvest fruit losses remain a persistent challenge, with estimates ranging from 20% to 27% of total production being lost before reaching consumers [1], [2]. Among the different stages of the supply chain, transportation contributes disproportionately to these inefficiencies. For example, a study on the banana value chain revealed that 7.6–8% of total losses occur solely during the movement of bananas between wholesale and retail stages, largely due to poor handling and inadequate transport methods [2].

The root causes of transport-related spoilage are well documented. Farmers frequently rely on open trucks and tractors, where fruits are bulk loaded without adequate protective packaging. This leads to mechanical damage from compression, vibration, and exposure to sunlight, which accelerates spoilage [3], [4]. Moreover, Sri Lanka's cold-chain infrastructure remains limited, especially for smallholder farmers, who dominate fruit production. Unlike large exporters, smallholders often cannot access refrigerated vehicles or standardized packaging solutions, leaving them vulnerable to high levels of wastage [6].

Attempts have been made to mitigate these issues. The National Institute of Post-Harvest Management (NIPHM) has introduced initiatives such as the distribution of plastic crates and farmer training programs, but these interventions have not been adopted at scale [5]. As a result, the absence of standardized packaging, weak coordination, and limited cold-chain logistics

continue to undermine efficiency, while mismatched vehicle assignment to perishable produce worsens spoilage [4].

Globally, researchers have proposed logistics optimization frameworks to address spoilage in perishable supply chains. The Vehicle Routing Problem (VRP) and its extensions, such as the Perishable Inventory Routing Problem (PIRP), introduce perishability constraints, delivery time windows, and heterogeneous fleets to improve both efficiency and freshness preservation [7]. In addition, dynamic routing algorithms, including Dijkstra's and A*, have been applied to minimize transit delays and reduce freshness decay [8].

However, these models have primarily been designed for large-scale, export-oriented supply chains in developed economies, where cold-chain infrastructure is already established [7]. In developing contexts like Sri Lanka, inefficiencies stem not only from routing but also from systemic challenges such as poor cold storage availability, lack of quality monitoring, and limited coordination among farmers, transporters, and distributors [4], [6]. This makes direct application of global models impractical without contextual adaptation.

In summary, the literature highlights that while Sri Lanka has quantified post-harvest losses and introduced small-scale packaging interventions, there is still no integrated solution that dynamically assigns vehicles based on perishability, optimizes routing, and incorporates real-time transport monitoring. This gap underscores the need for a localized, intelligent logistics framework tailored to Sri Lanka's fruit supply chain.

1.2 Literature Survey

Sri Lanka's fruit supply chains exhibit persistent post-harvest losses that multiple studies have quantified at national and commodity levels. Country overviews report 20–27 percent losses across fruits before retail, driven largely by weaknesses during handling and transport [1]. A detailed life-cycle assessment of the banana chain found an overall 27 percent loss, including 7.6–8 percent specifically between wholesale and retail due to in-transit damage and delays [2]. These findings establish transportation as a disproportionately loss-intensive stage that merits targeted logistical intervention.The operational drivers of transport-related spoilage are well documented. Bulk loading in open tractors and trucks, inadequate cushioning, and exposure to heat and vibration cause compression injury and accelerated deterioration, especially for climacteric fruits such as banana and mango [3], [4]. Packaging gaps are central: sector briefings identify insufficient crate adoption and poor unitization as major contributors to mechanical damage and stack collapse during haulage [4]. Although the National Institute of Post-Harvest Management has promoted plastic crate distribution and training, diffusion remains uneven and below the level needed for system-wide impact [5].

Institutional and structural factors amplify these technical weaknesses. Smallholder farmers dominate Sri Lanka's fruit production and typically rely on informal, opportunistic transport arrangements with limited access to cold-chain capacity or standardized handling protocols [6]. This fragmentation reduces the feasibility of coordinated backhauls, leads to mismatches between vehicle capacity and load characteristics, and constrains monitoring or accountability for en-route quality loss [3], [6]. In parallel, the global literature has advanced optimization approaches that directly link logistics decisions to perishability. Vehicle Routing Problem formulations with freshness decay, heterogeneous fleets, and time windows aim to minimize cost while preserving quality, and Perishable Inventory Routing Problem models integrate replenishment with delivery under shelf-life constraints [7]. Dynamic shortest-path and A*-family methods have been embedded within multiobjective frameworks to trade off distance, time, and freshness, sometimes incorporating simple temperature profiles or service-time penalties to approximate quality loss over routes [8]. These models demonstrate that routing, scheduling, and vehicle selection can be jointly optimized to reduce spoilage.

However, most implementations assume baseline infrastructure such as reliable cold-chain assets, standardized packaging, and robust telemetry for condition monitoring. These assumptions rarely hold in smallholder-led contexts like Sri Lanka, where gaps in cooling, packaging compliance, and data availability limit direct transfer of global templates [1], [4]-[6]. As a result, the literature points toward the need for localized designs that pair optimization with pragmatic constraints, for example, selectively allocating refrigerated vehicles to the most perishable loads, embedding packaging requirements within assignment rules, and leveraging lightweight telemetry for temperature and delay flags. Recent Sri Lankan discussions on the "hidden middle" in horticultural chains underscore coordination problems across collection, wholesale, and retail interfaces, where handoffs are frequent and accountability for quality loss is diffuse [3]. This further supports integrating assignment, routing, and basic in-transit monitoring into a single operational layer that can function amid variable infrastructure. Evidence from crate programs and training initiatives also suggests that packaging standards can be codified into logistics decision rules to reduce compression and vibration damage during haulage [4], [5].

In summary, the literature establishes three pillars for an actionable solution in Sri Lanka: first, transport is a high-leverage stage for reducing fruit losses [1], [2]; second, packaging and handling deficits are material contributors that must be operationalized within logistics rules [4], [5]; third, global optimization methods offer a foundation but require adaptation to smallholder and limited-infrastructure realities. These insights motivate a context-sensitive logistics approach that aligns vehicle assignment and routing with perishability and embeds minimum packaging and condition checks into routine transport operations.

1.3 Research Gap

Existing studies and initiatives in Sri Lanka's fruit and vegetable supply chain highlight significant progress in documenting postharvest losses, experimenting with marketplaces, and deploying corporate-level cold-chain infrastructure. Research from SLTC has quantified banana losses at different supply chain stages, confirming that poor handling and long transit times are major causes of waste [8]. Platforms like Helaviru and student-led concepts such as AgroLink have explored online trading, while large corporations such as CIC Agribusiness and projects like the LAPMC cold-chain center have implemented infrastructure-heavy solutions for their own networks. Similarly, training programs by the National Institute of Post-Harvest Management (NIPHM) and farmer registries from the Department of Agriculture support awareness and stakeholder identification. However, several critical research and practice gaps remain, especially when considering practical, scalable solutions for Sri Lanka's fragmented, smallholder-driven supply chains.

One major gap is that most interventions do not directly link perishability awareness with vehicle assignment or routing decisions. While perishability is well-documented in academic studies, there is no system that automatically matches crop batches to suitable vehicles (e.g., refrigerated vs. non-refrigerated) based on perishability, load size, and available fleet. This leaves smallholders especially vulnerable to spoilage when mixing consignments. Another gap is the lack of real-time, sensor-driven logistics adaptation. Current systems either study losses retrospectively or rely on fixed cold-chain corridors. None of the reviewed platforms support live re-routing when conditions such as temperature breaches, humidity spikes, or traffic delays occur. Without this, spoilage risks remain unmitigated even after initial planning.

A third gap concerns IoT-enabled, end-to-end monitoring at scale. Corporate actors like CIC may use internal sensors, but affordable, distributed IoT solutions for smallholders are missing. This prevents early detection of risks and limits traceability, especially for mixed consignments originating from different farmers. There is also a gap in integrating marketplaces with logistics execution. Helaviru and similar platforms allow produce trading, but logistics remains decoupled. Farmers must rely on intermediaries or arrange transport independently, which introduces delays, costs, and potential trust issues. A combined trade–transport–settlement workflow has not yet been implemented in Sri Lanka.

Another important gap lies in smallholder accessibility and affordability. Most existing solutions either require continuous internet, corporate membership, or infrastructure investment, which excludes the majority of rural farmers. Offline-first, cooperative pooling, and pay-per-use pricing models are not yet integrated into logistics solutions, despite being essential for adoption in Sri Lanka's rural context. Finally, evaluation and impact measurement is underdeveloped. While loss percentages and adoption statistics are reported, few controlled field pilots demonstrate how integrated digital logistics reduce waste or improve farmer income. Without such validation, claims about effectiveness remain largely theoretical.

In summary, current literature and initiatives provide valuable partial solutions such as cold storage, online marketplaces, and loss documentation but they remain fragmented and disconnected from real-time, end-to-end logistics optimization. The proposed FreshRoute system addresses these gaps by combining perishability-aware vehicle assignment, real-time sensor-driven re-routing, affordable IoT kits, integrated trade–logistics workflows, and smallholder-inclusive mobile access. This integrated approach fills the research gap by offering a practical, scalable, and locally adapted solution to reduce postharvest waste in Sri Lanka's fruit supply chain.

Feature / Gap	[9]	[10]	[11]	[12]	[13]	[14]	Proposed Solution
Perishability-aware vehicle assignment	No	No	No	No	Partial	No	Yes
End-to-end IoT & condition monitoring	No	No	No	No	No	No	Yes
Marketplace-Logistics integration	No	No	Yes	Yes	No	No	Yes
Packaging & handling protocols	Partial	Partial	Partial	Yes	No	Yes	Yes
Smallholder affordability, inclusion & offline UX	No	No	Partial	No	No	No	Yes

Table 1: Research gap comparison table.

1.4 RESEARCH PROBLEM

The primary research problem addressed in this study revolves around the pressing issue of post-harvest fruit spoilage in Sri Lanka's agricultural supply chain and the urgent need to mitigate this challenge through the adoption of intelligent logistics management practices. Despite Sri Lanka's rich agricultural diversity and year-round fruit production, a significant proportion of produce never reaches consumers in optimal condition due to inefficiencies in handling, transportation, and distribution. This study, therefore, seeks to explore how vehicles can be dynamically allocated and scheduled according to the perishability levels of different fruits, ensuring that highly perishable items receive priority in transportation. In addition, it investigates how transportation routes can be intelligently optimized, in order to reduce unnecessary delays that contribute to spoilage. Furthermore, the research aims to examine how real-time monitoring technologies and integrated handling practices can be implemented to improve traceability, enhance coordination across stakeholders, and maintain quality standards throughout the supply chain. The overarching challenge lies in designing and developing a context-sensitive, data-driven system that not only responds to dynamic variables such as fruit type, packaging requirements, and local infrastructure limitations but also ensures that fresh produce reaches wholesale and retail markets in the best possible condition. Ultimately, addressing this research problem holds the potential to reduce food waste, improve farmer incomes by eliminating inefficiencies, and contribute to a more sustainable and resilient food distribution system in Sri Lanka.

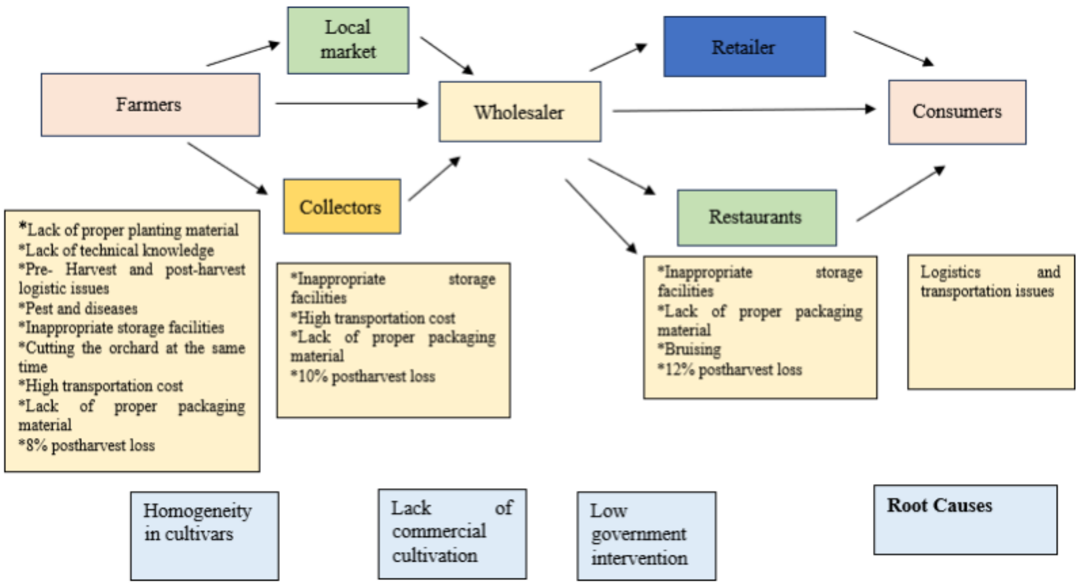

Figure 3: Overview of fruit supply chain

Figure 4: Poor transportation practices

1.5 RESEARCH QUESTIONS

Primary Research Question

- How can a spoilage-aware logistics system that integrates vehicle allocation, optimized routing, and transport condition monitoring reduce post-harvest fruit losses in Sri Lanka's supply chain?

Secondary Research Questions

- To what extent can dynamic vehicle assignment, based on fruit type, perishability, and storage needs, minimize spoilage compared to current manual allocation practices?
- How effective are routing algorithms in reducing transport delays and freshness decay in short-haul fruit logistics within Sri Lanka?
- What role can real-time sensor data (temperature, humidity, GPS tracking) play in improving transparency and timely interventions during fruit transport?
- How can a mobile-based booking and monitoring platform be designed to ensure accessibility for smallholder farmers with limited digital literacy?
- What are the infrastructural and cost-related challenges in implementing a spoilage-reduction logistics framework in rural Sri Lankan contexts?
- What is the estimated reduction in post-harvest losses (percentage terms) when the proposed system is compared against existing manual, paper-based logistics methods?
- How can the adoption of such a system impact farmer incomes, transporter efficiency, and overall supply chain sustainability in Sri Lanka?

2 RESEARCH OBJECTIVES

2.1 Main Objective

The main objective of this research is to develop a comprehensive, intelligent logistics system for the Sri Lankan fruit supply chain, specifically aimed at minimizing post-harvest losses through spoilage-aware transport management. The system, referred to as the "Smart Fruit Supply Chain System," will integrate cutting-edge data-driven decision-making methods, dynamic vehicle assignment, and route optimization algorithms tailored to the perishability of different fruits. The study focuses on addressing critical inefficiencies in transport, handling, and storage by combining real-time spoilage monitoring with optimized logistics strategies. The system will evaluate the perishability, type, and packaging requirements of fruits to determine the most suitable vehicle for collection and delivery, ensuring minimal quality degradation during transit. In addition, the proposed platform will offer a fully integrated approach that coordinates vehicle assignment, packaging considerations, and route optimization, providing a holistic view of the supply chain. By leveraging these advanced technologies, the study aims to reduce waste, improve the reliability of deliveries, and support data-driven decision-making for farmers, transporters, and other stakeholders. Ultimately, the research seeks to provide a context-sensitive, intelligent solution capable of transforming fruit supply logistics in Sri Lanka, enhancing both economic and operational outcomes for the agricultural sector.

2.2 Sub Objectives

Sub Objective 1: To Quantify Post-Harvest Losses and Identify Critical Spoilage Points

This objective focuses on measuring fruit losses at various stages of the supply chain, including collection, transportation, and delivery to markets. The goal is to understand where spoilage occurs most frequently and under what conditions, forming a basis for designing targeted interventions. Accurate quantification will help prioritize strategies for vehicle-perishability matching, route optimization, and handling improvements.

Sub Objective 2: To Design a Vehicle Assignment Mechanism Based on Perishability

This sub-objective aims to develop a dynamic system for assigning vehicles that considers the type of fruit, its perishability, and specific handling requirements such as temperature control and packaging. The system will ensure that fruits with higher spoilage risk are transported under optimal conditions, reducing quality degradation and economic losses.

Sub Objective 3: To Implement Dynamic Routing Optimization

The study will develop routing algorithms that account for perishability constraints, transport time, and local road conditions. By integrating spoilage risk into route planning, the system will ensure that produce reaches its destination in the shortest and safest possible time, reducing losses while increasing efficiency.

Sub Objective 4: To Integrate Real-Time Monitoring of Transportation and Handling

This objective involves developing mechanisms for real-time monitoring of environmental conditions, such as temperature, humidity, and handling practices during transport. The system will use this data to adjust routing and vehicle assignments dynamically, ensuring optimal conditions throughout the supply chain.

3 METHODOLOGY

The development of the Smart Fruit Supply Chain System follows a structured process to create an integrated, intelligent platform for minimizing post-harvest fruit spoilage in Sri Lanka. The first phase involves collecting high-quality historical and real-time data from farms, including fruit type, harvest quantity, perishability, transport availability, and environmental conditions such as temperature and humidity. This data will undergo pre-processing to ensure consistency, including normalization, handling missing values, and formatting GPS and sensor readings for algorithmic use. Following data preparation, the system will implement the Vehicle Assignment Module, which matches vehicles to fruit types based on perishability, storage requirements, and handling conditions. Concurrently, the Routing Optimization Module will be developed using algorithms to calculate optimal transport routes, incorporating spoilage risk, transit time, and vehicle capacity. These modules will be integrated with a Monitoring and Reporting Module, which tracks the status of transports in real-time, records environmental data, and generates actionable insights for farmers and transporters. Once the modules are implemented, iterative testing will be performed using simulated and real-world scenarios to evaluate spoilage reduction, on-time delivery, and operational efficiency. Finally, an intuitive user interface will be developed to allow farmers and logistics providers to easily access vehicle assignment, routing recommendations, and transport monitoring dashboards. The fully integrated system aims to deliver a comprehensive solution, combining perishability-aware vehicle assignment, dynamic routing, and real-time monitoring to enhance efficiency, reduce post-harvest losses, and improve profitability for stakeholders across the fruit supply chain.

3.1 System Architecture Diagram

The system architecture of the Smart Fruit Supply Chain System is designed to seamlessly integrate farmers, transporters, distributors, and IoT sensors through a centralized web application. Farmers provide harvest data, transporters share vehicle availability and details, and distributors submit delivery requirements, while IoT sensors continuously supply environmental and product-related data. All these inputs are funneled into the Web Application Interface, which acts as the central access point. The interface connects directly to the Web Application running on the Web Server, where key modules such as the Vehicle Assignment Module, Routing Optimization Module, and Monitoring and Reporting Module operate collaboratively.

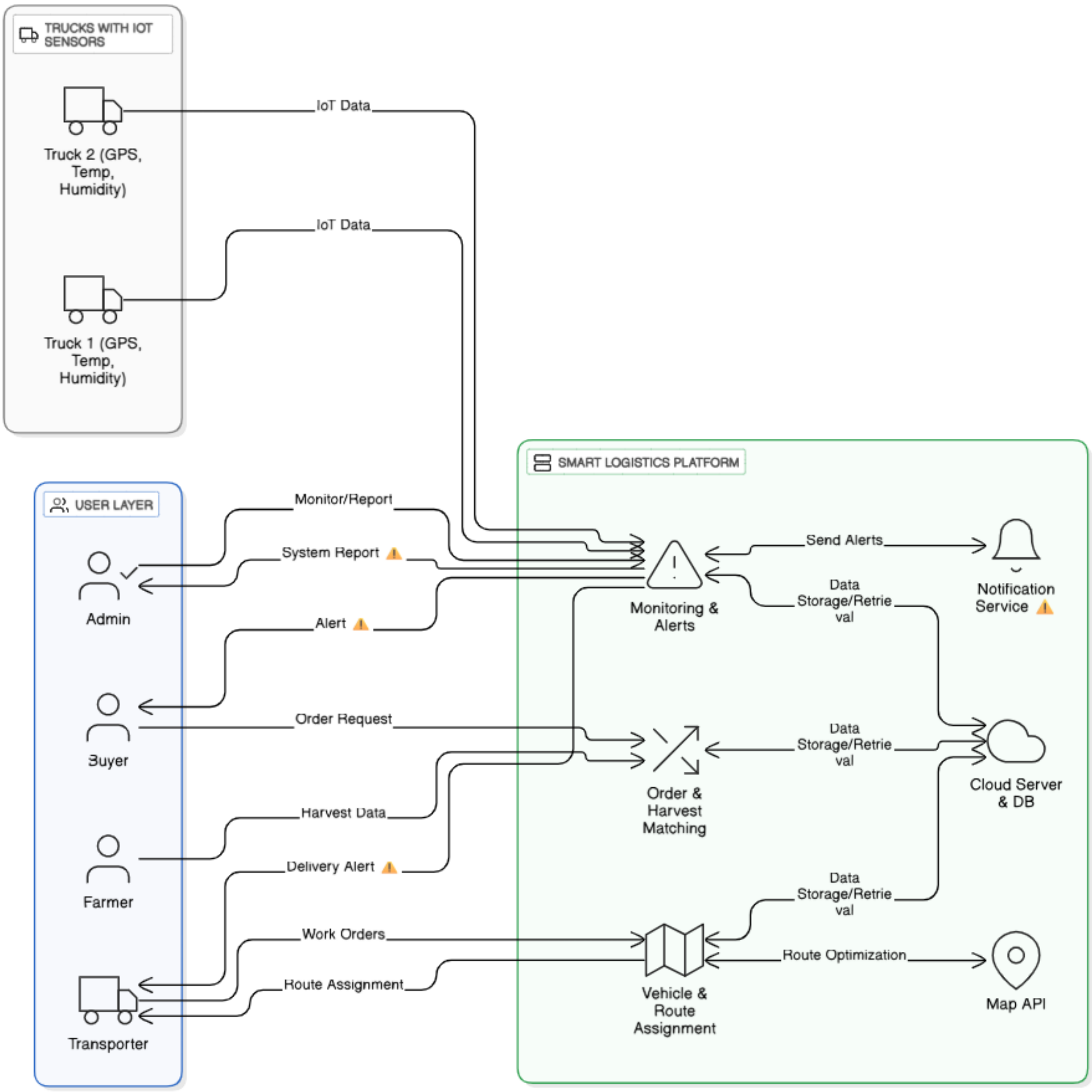

Figure 5: System architectural diagram

These modules process data to dynamically assign vehicles, optimize transport routes, and enable real-time tracking and reporting. The Web Server further integrates with a cloud-based database that stores transport records, spoilage patterns, and other system data, ensuring reliable storage and retrieval of information. This structured flow ensures efficient decision-making, minimizes fruit spoilage, reduces dependency on middlemen, and creates a transparent, data-driven fruit supply chain in Sri Lanka.

4 PROJECT REQUIREMENTS

4.1 Functional Requirements

• The system must match buyer orders with available farmer harvest data dynamically.

• Vehicles should be automatically assigned based on order quantity, cooling needs, and proximity to farmers.

• The system shall optimize delivery routes using algorithms enhanced with perishability factors and delivery deadlines.

• IoT sensors on vehicles must continuously monitor temperature, humidity, and GPS location.

• The system must send alerts to transporters, buyers, and admins if any cold-chain deviation or delay occurs.

• Delivery confirmation and status updates must be recorded in the central database.

• The system shall generate alerts when sensor readings indicate conditions that may lead to spoilage.

• The system shall provide farmers with dashboards to view vehicle assignments, expected pickup times, and delivery updates.

• The system shall provide transporters with optimized route plans, the ability to update delivery status, and options to report handling or vehicle issues.

• The system shall provide distributors and buyers with dashboards to monitor incoming supply, fruit quality status, and estimated arrival times.

4.2 Non-Functional Requirements

1.Reliability

The Smart Fruit Supply Chain System is intended to be used by farmers, transporters, and buyers for critical decision-making related to harvest scheduling, vehicle allocation, and quality assurance. Hence, the reliability of the system must be very high in order to minimize post-harvest losses and ensure trust among stakeholders. Any inaccurate routing, forecasting, or quality detection may lead to economic losses and spoilage. To guarantee reliability, the system will undergo extensive field testing with real transport and market data, along with simulation experiments to validate the consistency and accuracy of its recommendations.

2.Security

The system handles sensitive information, including farmer harvest details, transaction records, and buyer contracts. Security of this data is of paramount importance to prevent fraud, data breaches, or unauthorized access. Strong encryption methods, secure authentication protocols, blockchain immutability, and regular security audits will be implemented to guarantee the privacy and protection of stakeholder data. Access will be role-based, ensuring that farmers, transporters, and buyers can only view relevant information.

3.Availability

Since agricultural operations and transportation run continuously, the system must be available at all times to support critical supply chain activities. High availability will be achieved through server redundancy, cloud hosting, and load balancing to prevent downtime. The system is designed to function in rural areas with poor connectivity by offering offline data entry and automatic synchronization when internet access is restored.

4.Usability

The system will provide a user-friendly interface accessible via mobile and web platforms, allowing farmers, transporters, and buyers to interact with features easily. The design will support Sinhala, Tamil, and English languages to ensure inclusivity. Dashboards will be intuitive, offering clear visualizations of routes, vehicle assignments, and spoilage monitoring data. Usability testing will be conducted with end-users to ensure minimal training is required to operate the system effectively.

5.Scalability

As adoption of the system expands across Sri Lanka's fruit supply chain, it must be able to scale seamlessly to handle large numbers of farmers, transporters, and buyers. The architecture will be cloud-enabled to support increasing data volumes, transactions, and connected devices (e.g., IoT sensors). The system will be optimized to process real-time data streams without performance degradation, even during peak harvest seasons when network traffic is high.

6.Performance

The system must deliver fast response times for decision-making. Vehicle assignment and routing recommendations should be generated within a few seconds, while demand forecasts and quality assessment results must be provided in near real-time to ensure timely harvesting and transport. Sensor data and GPS updates will be processed with minimal latency, enabling accurate and actionable monitoring. Optimization techniques will be applied to ensure that performance standards are maintained across varying workloads.

4.3 Personal Requirements

Expert support in fruit supply chain management

Interdisciplinary collaboration is crucial for the development of the Smart Fruit Supply Chain System, requiring expertise in agricultural production, post-harvest handling, and logistics management. Guidance from specialists with in-depth knowledge of fruit cultivation, harvest planning, and storage practices is essential to design a system that reflects real-world supply chain scenarios. These experts provide insights into factors such as fruit perishability, packaging requirements, and transportation constraints, ensuring that the system effectively reduces post-harvest losses. Strong supervision from domain experts will be necessary to maintain high standards in data collection, operational planning, and validation of transport and handling practices.

Deep understanding of the intersection between agriculture and logistics

The success of this research depends on comprehensive knowledge of fruit supply chains and efficient logistics management. Understanding the characteristics of different fruit types, their perishability, and optimal post-harvest handling is essential for planning collection, transport, and storage strategies. This expertise ensures that the system addresses practical challenges in the supply chain, minimizes spoilage, improves farmer incomes, and enables efficient coordination among farmers, transporters, and distributors. By combining knowledge of agricultural production and logistics, the system can deliver actionable insights that strengthen Sri Lanka's fruit supply chain and enhance overall operational efficiency.

4.4 Use Cases

1.Vehicle Assignment

Actor: System
Description: The system automatically assigns vehicles to fruit batches based on perishability levels, load capacity, temperature requirements. The allocation logic ensures that fruits with shorter shelf lives are prioritized for faster and safer transport.
Outcome: High-risk fruits are delivered in optimal conditions, reducing spoilage and ensuring freshness upon arrival.

2.Real-Time Delivery Tracking

Actor: Distributor
Description: Distributors access a live tracking dashboard that displays the shipment's location, estimated time of arrival, and alerts related to delays or potential spoilage risks. Notifications are automatically triggered if thresholds for temperature or handling conditions are breached.
Outcome: Distributors are able to prepare storage facilities in advance and take proactive measures to avoid product loss.

3.Perishability Alerts & Notifications

Actor: Farmers / Transporters
Description: The system sends automated alerts when fruits nearing critical perishability thresholds are detected in storage or during transit. Notifications can be sent to farmers to speed up dispatch or to transporters to prioritize delivery.
Outcome: Timely interventions are made to prevent wastage and maximize freshness.

4.5 Expected Test Cases

Rigorous testing for accuracy, reliability, usability, and scalability will be performed in the development and implementation of the Fruit Supply Chain Management System. Below are the key test cases expected for this research project.

i.

Objective	To verify that the system correctly assigns vehicles based on fruit perishability, load capacity, and handling requirements.
Test Case	Input fruit batches with varying perishability (short shelf life vs. long shelf life) and vehicle constraints (capacity, refrigeration). Verify that high-risk fruits are prioritized for faster and safer transport.
Expected Outcome	The system should allocate vehicles optimally without errors, ensuring perishable fruits are matched with appropriate transport.

Table 2: Test case 1

ii.

Objective	To assess whether the system provides accurate real-time shipment tracking with live updates and spoilage risk alerts.
Test Case	Track a shipment under varying network conditions. Introduce delays and temperature breaches to test if alerts are triggered and delivery time estimates are updated.
Expected Outcome	The system must display accurate location, ETA, and risk notifications.

Table 3: Test case 2

iii.

Objective	To ensure that farmers, transporters, and distributors receive timely alerts regarding perishability thresholds, delays, or risks.
Test Case	Simulate scenarios where fruits are close to expiry, vehicles break down, or temperature thresholds are exceeded.
Expected Outcome	Alerts should be delivered instantly to relevant users via the system dashboard or notifications.

Table 4: Test case 3

iv.

Objective	To assess the usability of the web-based system for all stakeholders (farmers, transporters, distributors).
Test Case	Test navigation, data entry, report viewing, and notifications across devices (desktop, tablet, mobile). Gather user feedback on ease of use.
Expected Outcome	The application should be responsive, user-friendly, and require minimal training.

Table 5: Test case 4

v.

Objective	To test system response times and scalability under increased user and data load.
Test Case	Gradually increase the number of farmers, transporters, and distributors using the system simultaneously while uploading large datasets.
Expected Outcome	The system should maintain minimal response times and scale efficiently without crashes or slowdowns.

Table 6: Test case 5

5 COMMERCIALIZATION

The "Smart Fruit Supply Chain" project introduces an innovative AI-driven platform designed to transform the agricultural supply chain by ensuring fruit freshness, reducing spoilage, and optimizing logistics. Unlike conventional supply chain management systems, this solution leverages real-time sensor data, AI/ML models for spoilage prediction, and intelligent routing optimization to maximize efficiency from farm to distributor. By directly addressing the critical issues of post-harvest losses and inefficient transportation, this platform fills a significant gap in the agri-tech industry.

The commercialization strategy positions the solution for adoption across multiple stakeholders in the agricultural ecosystem, including farmers, transporters, distributors, retailers, and export companies. Through integration with web and mobile platforms, even small-scale farmers will be able to access insights that were previously only available to large-scale commercial farms. Exporters and distributors benefit from improved shelf-life monitoring and more reliable logistics operations, which directly enhance profitability and customer satisfaction. The system also opens opportunities for partnerships with agricultural cooperatives, logistics providers, and government-led smart farming initiatives, enabling large-scale implementation across regions. Over time, integration with blockchain-based traceability systems can strengthen transparency and trust in both local and global fruit markets.

To accelerate market penetration, the project will engage in targeted marketing campaigns, collaboration with agricultural boards, presentations at agri-tech exhibitions, and pilot projects with local farmer groups. Demonstrating measurable reductions in fruit wastage and improved profitability during these pilot phases will build trust and drive adoption at scale.

With the rising global demand for sustainable and efficient food supply chains, the "Smart Fruit Supply Chain" solution is well-positioned for commercialization, offering tangible economic and environmental benefits while contributing to global food security.

6 SOFTWARE SPECIFICATIONS

In the development, the Smart Fruit Supply Chain System will utilize a comprehensive set of software tools to ensure scalability, robustness, and efficiency. The specifications focus on delivering strong performance, seamless usability, and secure data handling. With these technologies, the system can effectively minimize post-harvest fruit spoilage and enhance overall supply chain efficiency.

Category	Technology	Purpose
Programming Languages	Python	Development of algorithms, data preprocessing, and backend logic.
	JavaScript	Frontend development for interactive web interface.
	HTML / CSS	Structuring and styling the web application.

Frameworks	TensorFlow / PyTorch	Model development for routing optimization and perishability analysis.
	Flask / Django	Backend development, API management, and server-side operations.
	React.js, React Native	Frontend development for user dashboards and mobile interfaces.
Libraries	Pandas / NumPy	Data manipulation and preprocessing.
	NetworkX	Graph-based routing algorithms (Dijkstra's, A*).
Database	PostgreSQL / MongoDB	Centralized storage of harvest data, transport records, and monitoring logs.
	Firebase	Real-time syncing of monitoring data.
APIs	REST API	Communication between frontend, backend, and database.
	Google Maps API	Integration of geolocation and routing data.
Development Environment	Visual Studio Code	Primary IDE for frontend and backend development.
	PyCharm	IDE for Python and ML model development.
	GitHub	Version control and collaborative development.
	Docker	Containerization for deployment and scalability.
Cloud / Hosting	AWS	Hosting, scalability, and secure deployment.

Table 7: Software Specifications

7 BUDGET

Description	Amount (LKR)
Web hosting	5000.00
IOT sensors	2000.00
Data collection, surveys, and field visits	5000.00
Mobile Development tools cost	10000.00
Internet cost	3000.00
Cloud services	5000.00
Miscellaneous	10000.00
Total Estimated Budget	40000.00

Table 8: Budget

The estimated budget of LKR 40,000 is allocated to cover the essential components required for the successful implementation of the proposed spoilage-reduction logistics system. Web hosting (LKR 5,000) is necessary to deploy the online platform accessible to farmers, transporters, and distributors. A small set of temperature and humidity sensors (LKR 2,000) will be used to capture real-time transport conditions, ensuring reliable monitoring of spoilage risks. Data collection, surveys, and field visits (LKR 5,000) are planned to gather accurate, localized insights from stakeholders in the fruit supply chain. Mobile development tools (LKR 10,000) are budgeted to build the farmer-friendly mobile application, while internet costs (LKR 3,000) will support research, collaboration, and testing. Cloud services (LKR 5,000) are essential for data storage, computation, and running optimization algorithms in real-time. A provision for miscellaneous expenses (LKR 10,000) has been included to cover unforeseen needs such as additional equipment, printing, or transport during fieldwork. Altogether, this budget ensures that both the technical development and practical validation of the system are feasible within the scope of the research.

8 WORK BREAKDOWN STRUCTURE

The work breakdown structure for the Smart Fruit Supply Chain System is defined through six major stages: initiation, planning, design, implementation, enhancement, and closure. The initiation phase establishes the foundation of the project by defining its objectives, identifying all key stakeholders including farmers, transporters, distributors, and retailers, and preparing essential documents such as the project charter and proposal. In the planning phase, the system requirements are analyzed, and feasibility studies are carried out to ensure technical and operational viability. A detailed project plan is developed, including risk assessment, timelines, and milestones, to guide the development process and ensure successful delivery. The design phase focuses on creating the overall system architecture, designing databases, and defining integration strategies for mobile, web, and IoT components. It also includes designing modules such as the farmer app, transporter app, distributor dashboard, and admin portal, ensuring that the system is scalable, secure, and user-friendly. The implementation phase translates the designs into working solutions. This includes developing and deploying the mobile and web applications, integrating IoT and GPS tracking for transport monitoring, and implementing logistics optimization algorithms for route and vehicle assignment. Rigorous unit and integration testing validate the system's performance and reliability.

Finally, the closure phase wraps up the project by conducting a comprehensive project review, preparing the final report, and publishing results or findings for wider use in industry or academia. Training sessions and recommendations for future improvements, such as scaling to new regions or integrating AI-driven analytics, are also delivered to ensure long-term sustainability. Each phase of this work breakdown structure ensures that the Smart Fruit Supply Chain System is systematically planned, implemented, and enhanced to meet the real-world needs of farmers, distributors, and consumers effectively.

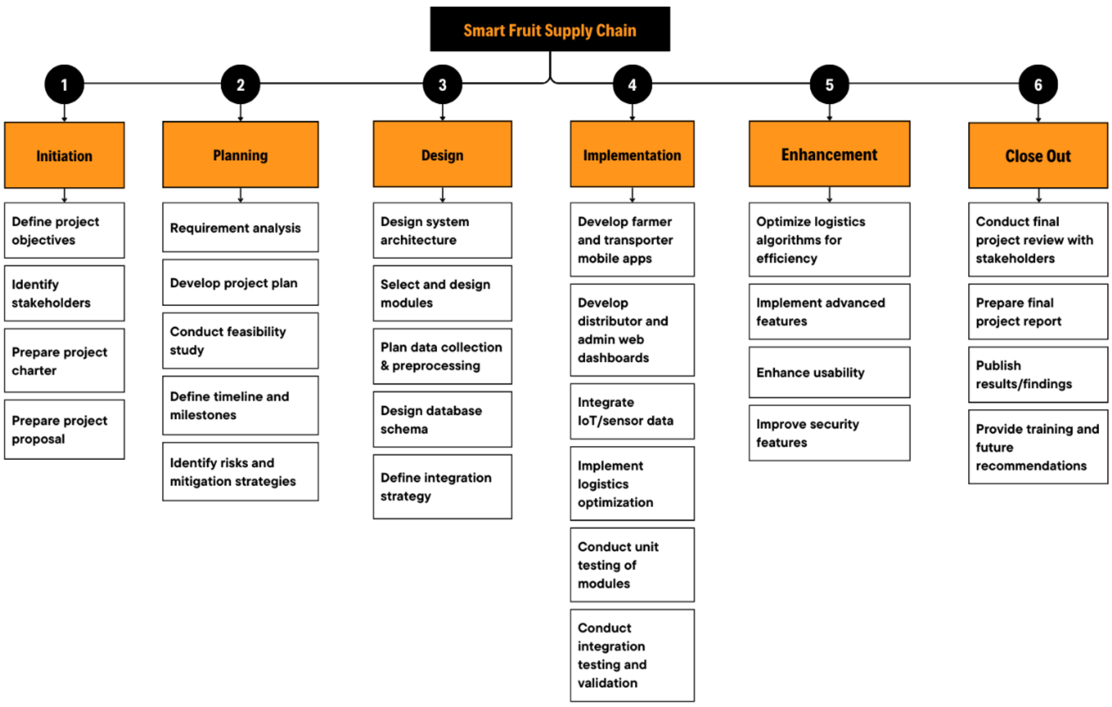

Figure 6: Work Breakdown Structure

9 GANTT CHART

The Gantt chart outlines the project timeline from May 2025 to May 2026, covering all major phases of the research lifecycle. The process begins with topic selection, supervisor registration, feasibility study, and requirement gathering, followed by literature review and solution design. From late 2025, the focus shifts to data collection, algorithm for routing optimization. The subsequent stages include backend and frontend implementation, integration, and multiple rounds of testing to ensure system reliability. Finally, the research paper preparation, reporting, and progress presentations lead up to the final presentation and viva in May 2026. This structured timeline provides a clear and systematic roadmap for completing the research successfully.

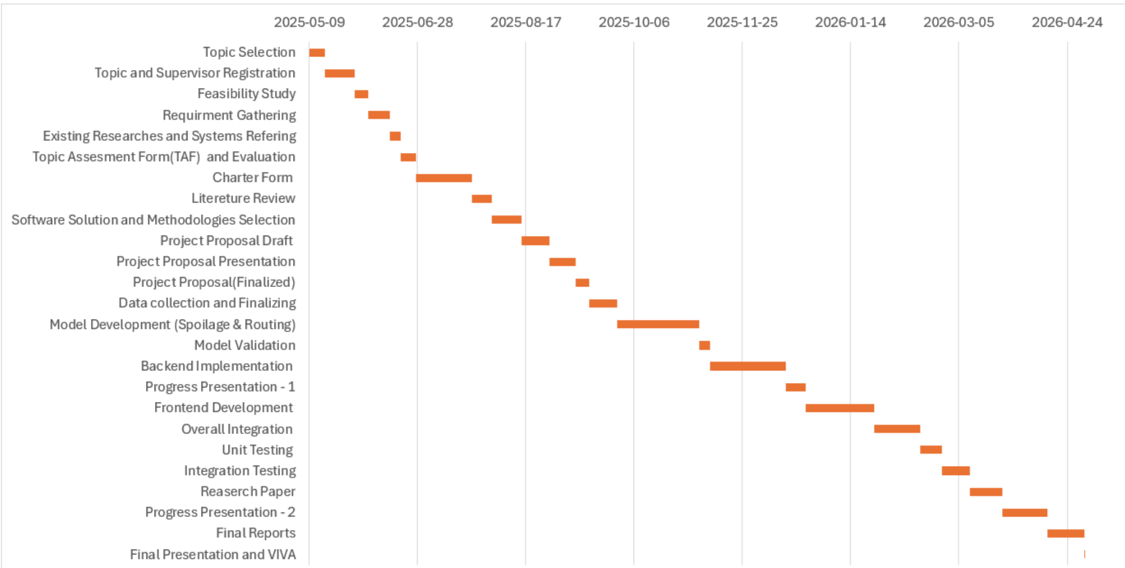

Figure 7: Gantt Chart

10 CONCLUSION

The proposed system for spoilage prediction and routing optimization represents a significant step forward in strengthening food supply chain management, particularly within perishable goods logistics. By integrating predictive analytics with intelligent route planning, the system directly addresses critical challenges such as excessive food waste, delayed deliveries, and inefficient resource allocation—issues that currently undermine both profitability and sustainability in the agricultural and food distribution sectors. Through spoilage prediction, stakeholders gain real-time insights into the condition and shelf life of products, enabling proactive decision-making such as rerouting goods to closer markets, prioritizing high-risk consignments, or adjusting storage and transport conditions. Coupled with dynamic routing optimization, the system ensures that perishable goods reach their destinations within optimal timeframes, minimizing deterioration and safeguarding food quality. This not only reduces financial losses but also enhances consumer trust in food freshness and safety.

Beyond immediate operational benefits, the system provides a scalable and adaptable framework applicable to multiple levels of the supply chain—from farm to retail. Its data-driven approach lays the groundwork for long-term improvements in logistics planning, demand forecasting, and inventory management, making it suitable for diverse agricultural products and different geographical contexts. Moreover, the proposed solution contributes directly to global sustainability goals by reducing food wastage, lowering carbon emissions from inefficient transportation, and improving the utilization of natural and financial resources. By empowering producers, distributors, and retailers with actionable intelligence, it fosters a more resilient supply chain ecosystem that can adapt to challenges such as climate change, market fluctuations, and increasing consumer demand for transparency.

In essence, this project goes beyond solving logistical inefficiencies; it creates a holistic pathway toward sustainable, cost-effective, and consumer-focused food supply chain management. Its successful implementation will not only benefit immediate stakeholders but also generate widespread socio-economic and environmental advantages, contributing to food security and improved quality of life for communities at large.

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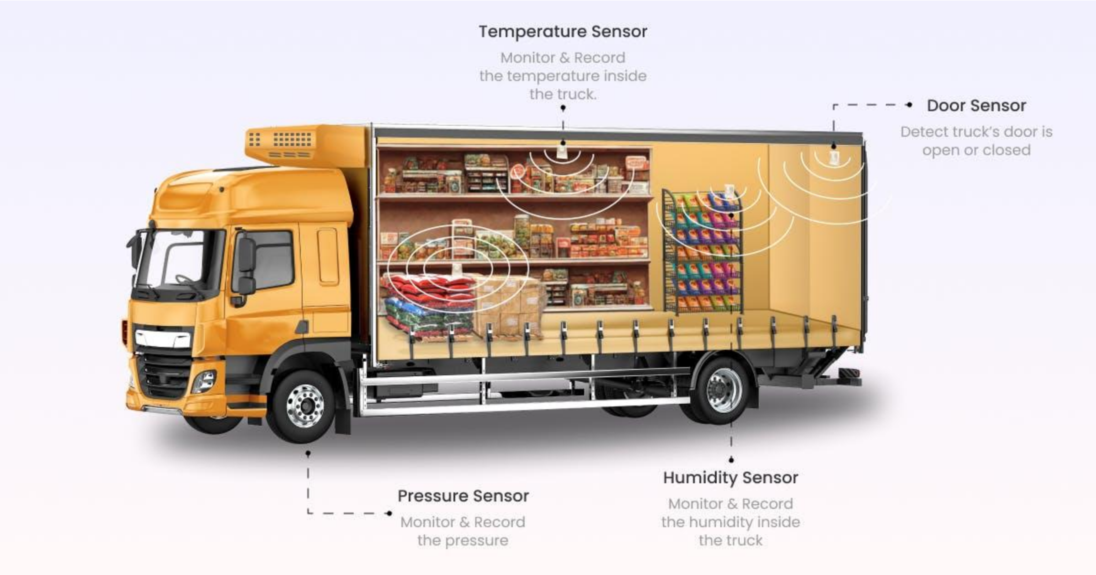
12 APPENDIX


Temperature Sensor
Monitor & Record the temperature inside the truck.
Door Sensor
Detect truck's door is open or closed
Pressure Sensor
Monitor & Record the pressure
Humidity Sensor
Monitor & Record the humidity inside the truck